Unit 21: Emerging Technologies.
Final Assignment Extended Brief (ver. 2.1)

Context

Understanding emerging technologies is important for sustainable professional development, especially in IT. Many emerging technologies are now at the intersection of traditional industries and digital solutions.

However, the national context is also important. The same technologies can be adopted differently depending on historical, economic, and cultural factors.

As future specialists who will contribute to the development of Uzbekistan, students must not only understand global trends but also be able to adapt technologies to local conditions and choose the most appropriate solutions for the local market.

Before preparing the final assignment for Unit 21: Emerging Technologies, each student must choose one emerging technology related to their IT specialization.

Important rule: choose the right technology

You must NOT choose:

- very general topics (AI, Internet, Computers)
- products (Flutter, ChatGPT, Telegram bot)
- old stable technologies (basic SQL, TCP/IP, OOP)

You must choose something:

- modern
- still growing
- used in real systems today
- with clear future impact

Approval step (mandatory)

Before starting the assignment, the student must:

- write a short description (100–150 words): what the technology is and why it is emerging.
- get approval from the teacher

You can use this text as the introductory part (Task 1) of your assignment. Without approval, the work will NOT be accepted.

Tasks

Task 1. Introduction

- Explain the idea of Emerging Technology
- Show key characteristics:
  - new / innovative
  - fast growth
  - high impact
- Explain why your chosen technology fits

Format: text 100-150 words (You can use your text from the approval procedure).

Task 2. Technology overview

- How it works (simple explanation)
- Where it is used today
- Real examples (companies, systems)

Advantages and disadvantages

- Benefits (speed, cost, automation, etc.)
- Problems (risks, limits, ethical issues)

Format: Three-part structured essay (see Appendix B for details).

Task 3. Economic factors

- Who created and supports this technology
- How companies make money
- Real business models

Format: Text 400-500 words (see Appendix C for details).

Task 4. Technology connections

You must create a diagram using Mermaid or another structured diagram tool. Show:

- how your technology connects to others
- what it replaces or improves

Format: One diagram for 20–25 blocks. (See Appendix D for more details.)

Task 5. Political and social factors

- Laws and regulations
- Impact on jobs, privacy, society

Format: Text 400-500 words + PEST analysis with diagram. You must include at least two **Uzbekistan examples**

Task 6. Conclusion - Future impact

- Will this technology grow or fail?
- What risks exist?
- Your personal opinion with arguments

Format: Text 100-150 words.

Task 7. Presentation

You will prepare a presentation (5-6 slides) based on Tasks 1 - 6. The presentation must show key aspects of chosen technology.

Appendix A. Checklist of evidence required

Anti-AI requirements (very important)

You can use AI tools to research, check, and translate your assignment. However, your work **must include original and personal elements**.

What is NOT acceptable:

- Copy-paste text from AI
- Very general AI text without examples
- AI generated diagrams
- No local context

Mandatory requirements:

1. **Local context (Uzbekistan)**
   - At least 3-4 real examples from Uzbekistan
   - If no direct example → explain why
2. **Diagram**
   - Must be created by YOU
   - Must match your explanation
3. **Personal choice explanation**
   - Why YOU selected this technology
   - Connection to your future job
4. **Comparison or evaluation**
   - Not only description
   - You must compare or judge something
5. **File formats**
   - All final files must be in PDF format.
   - All model calculations must have sources in Excel or Google Sheet format.
   - All diagrams must have sources in common editable formats: Draw.io, SVG, Mermaid, Miro, etc.

Task requirements:

Task 1. Introduction

- 100–150 word text
- Clear explanation of Emerging Technology concept
- Justification of chosen technology

Task 2. Technology overview and analysis

- Three-part structured essay
- Clear explanation of how the technology works and where it is used
- Real examples (companies, systems)
- Balanced advantages and disadvantages
- **Local Uzbekistan context:** include 2–3 real local cases (business usage, implementations, or realistic future predictions) using local laws, articles, or official sites

Task 3. Economic factors

- 400–500 word text
- Real companies and business models
- Explanation of how value is created (e.g. subscription, platform, API)
- 2–3 trusted sources
- **Local Uzbekistan context:** include 2–3 local business cases or market examples (real or justified future scenarios), with references to local sources

Task 4. Technology connections diagram

- One diagram (20–25 blocks)
- Created using Mermaid, Draw.IO, Miro or another structured tool
- Shows connections with other technologies
- Matches explanation in the report

Task 5. Political and social factors

- 400–500 word text
- One supporting diagram (e.g. stakeholder map, cause-effect, or similar)
- Analysis of laws, regulations, and social impact (PEST method)
- **Local Uzbekistan context:** use local laws, government programs, news, or official sources; include 2–3 cases or examples

Task 6. Conclusion – future impact

- 100–150 word text
- Clear personal evaluation
- Arguments and risks included

Task 7. Presentation

- 5–6 slides
- Covers key aspects of Tasks 1–6
- Clear and structured
- Be ready to present your technology to your classmates

Appendix B. Three-part structured essay.

In this task, you write a structured essay about your chosen technology. Use the three-part essay model. Think about it like software architecture. Clear structure helps your ideas work correctly.

Start with a hook. Show a real problem or situation from daily life or IT work. Then **write your main thesis**. This is your key idea about the technology and its importance.

Next, **create three theses:**

Thesis 1: how the technology works
Thesis 2: where it is used today
Thesis 3: advantages and problems

For each thesis, add 3 arguments. Arguments must be simple, practical, and based on real examples. Use companies, systems, or services. If possible, include examples from Uzbekistan.

Important rule:

Each thesis = 1 clear sentence
Each argument = 1 clear sentence

After that, **write a conclusion.** Summarize your three theses in a short form. **Finish with a future perspective**. Explain what may change in the next 5–10 years.

Keep your text simple and logical. Do not write long or complex sentences. The table below illustrates the three-part essay structure.

<table>
<tr><td colspan="3">Introduction (Intrigue / Hook)</td></tr>
<tr><td colspan="3">General Idea</td></tr>
<tr><td colspan="3">Thesis 1
Thesis 2
Thesis 3</td></tr>
<tr><td>Thesis 1 (repeat)</td><td>Thesis 2 (repeat)</td><td>Thesis 3 (repeat)</td></tr>
<tr><td>Argument 1.1
Argument 1.2
Argument 1.3</td><td>Argument 2.1
Argument 2.2
Argument 2.3</td><td>Argument 3.1
Argument 3.2
Argument 3.3</td></tr>
<tr><td colspan="3">General Conclusion</td></tr>
<tr><td colspan="3">Conclusion from Thesis 1
Conclusion from Thesis 2
Conclusion from Thesis 3</td></tr>
<tr><td colspan="3">Sight to the future (perspective)</td></tr>
</table>

Appendix C: Impact of Emerging Technologies on Economic Factors

Write a **400–500 word structured text** about the economic side of your technology. Your goal is to show how this technology creates business value in real life.

Use:

- 1–2 international examples
- 2–3 examples from Uzbekistan (real or realistic future cases)
- 2–3 trusted sources

Write clearly. Avoid vague phrases like “very big” or “very fast”.

Important writing rules:

- Write clear and short paragraphs
- Use real examples, not theory
- Support ideas with sources
- Focus on logic, not description

1. Origin and Time to Market

Explain who created this technology and when it was developed.

Show the timeline of the following stages: idea, first product, and real business use. Was there a delay between these stages? Explain why, citing reasons such as technical limitations, high costs, an unready market, or other factors.

2. Business Model and Monetization

Explain how this technology generates revenue. Select the most appropriate business model, such as subscription, pay-per-use, platform, or SaaS.

Describe real value creation, such as direct payment, ads, data, and API usage. Also, explain which model will not work and why. Provide at least one real-world company example.

3. Market Size and Customers

- Describe the market: size (money or users), growth growing, stable, or saturated).
- Identify the customers: individuals, businesses, or the government. Explain who pays the most and why.

4. Costs and Profit

- Explain the main costs: development, infrastructure (cloud and servers), marketing, and support.
- Identify the biggest cost. Analyze unit economics, including revenue per user and cost to acquire users. Is the business profitable?

5. ROI and Scalability

- Explain the return on investment, whether it be fast or long-term.
- Analyze the scalability. Can it grow to accommodate millions of users?
- Describe what happens when the number of users increases: cost per user and system performance.

6. Risks, Competition and Market Impact

- Identify risks: technical, financial, and market.
- Analyze competition: How easy is market entry? Are there alternative technologies?
- Explain the market impact: Who wins and who loses?

Bridge to Task 4: Briefly show how your technology connects with other technologies.

7. Regulation and Local Market (Uzbekistan)

- Analyze the regulation. Does the government support or limit this technology?
- Explain the local challenges related to infrastructure, user behavior, and legal issues. Give two to three examples from Uzbekistan (real or realistic).
- Answer the following question: Is this technology realistic for Uzbekistan now?

Bridge to Task 5: Explain how the regulation connects to social and political factors (next task).

Appendix D. Roadmap: Creating a Technology Interconnection Diagram

Step 1. Use GPT to generate an interconnection essay

Upload to GPT:

- Technology description essay (from Task 2)
- Economic factors essay (from Task 3)

Prompt:

For my technology [tech name], based on the attached essays, write an essay explaining how this technology connects to other technologies.
Include:

- *which technologies it depends on*
- *which technologies it inherited from*
- *which technologies it improves*
- *which technologies it replaces*

Analyse historical development and economic factors. Size: about 1000 words.

Step 2. Manual critical check (very important)

You must:

- Check facts using Google, Wikipedia, etc.
- Remove weak or incorrect ideas.
- Add missing technologies.

Step 3. Use GPT to extract technologies

Prompt:

From this text, extract all mentioned technologies.
Create a clear list.
Group them into 3–4 logical categories (for example: infrastructure, data, applications, hardware).

Step 4. Manually improve the list

You must:

- remove irrelevant items
- add important missing technologies
- keep 20–25 technologies total

Step 5. Use GPT to define maturity level

Prompt:

For each technology, assign a maturity category:

- *Obsolete (not widely used now)*
- *Mature (widely used, stable standards)*
- *Emerging (growing fast)*
- *Future (in research or early stage)*

Show the result as a table.

You must check logic:

- no random classification
- use real examples

Step 6. Manually build the diagram

Use Miro, Draw.io, MS Visio, Mermaid, or other business diagram software.

Follow rules from your activity :

- Use four color-coded blocks to show obsolete, mature, emerging, and future technologies.
- Use three to four connection types. For example: dependency, combination, enhancement, or disruption.
- Add short labels to the connections: "Needs data," "Improves speed," "Replaces," etc.

See the example diagram in the shared folder.

You can also generate a Mermaid diagram using a GPT prompt. Be sure to check the syntax and manually fix the layout if needed.

Prompt: *Using this technology list and relationships, create a Mermaid diagram.*

Appendix E: Simplified Pest-Analysis Method

Key Concepts

Letter	Category	What to look for?
P	Political	Laws, government policy, military use, rules
E	Economic	Investments, markets, cost to start, economic situation
S	Social	Public needs, culture, education, user behavior
T	Technological	New ideas, infrastructure, compatibility, level of development

Important: We only look at *external* factors — things the company or technology author cannot change directly.

STEP-BY-STEP INSTRUCTIONS

STEP 1: Collect the factors

Go through each PEST letter. Choose 2–3 factors from each section that affect your technology.

Example (spark gap radio, 1895–1910):

Category	Factor
P	− Governments use radio for military control − Possible for Info Interception
E	− Investment in communication infrastructure − Growth of military technology industry
S	− New role of information in war − Faster communication between ships
T	− Marconi ship sets − Coastal wireless stations

STEP 2: Rate the importance of each factor

We use two parameters:

A) Factor Weight (A) — How important is it IN GENERAL for your field?

Rating	Weight Range	How to choose
🔴 Critical	0.15–0.20	Factor can "start" or "stop" the technology
🟡 Significant	0.08–0.14	Affects speed of use or size of impact
🟢 Background	0.01–0.07	Mentioned, but does not change key decisions

Rule: The sum of all weights must be == 1.0 (or 100%)

Example of weight distribution:

Factor	Weight (A)	Reason
Governments use radio for military control	0.20	Key driver of demand and funding
Possible for Info Interception	0.10	Important risk, but not blocking
Investment in communication infrastructure	0.05	Supporting, but not deciding factor
Growth of military technology industry	0.15	Creates a market
New role of information in war	0.25	Changes how the technology is used
Faster communication between ships	0.15	Practical value for the navy
Marconi ship sets	0.05	Technical detail, not a strategic factor
Coastal wireless stations	0.05	Infrastructure element

B) Strength of Influence (B) — How STRONGLY does the factor affect things in a specific period?

Score	Meaning	Example anchor
1	Almost no influence	Factor is mentioned in sources, but does not change decisions
2	Weak influence	Considered sometimes, not a priority
3	Moderate influence	Affects tactical decisions, needs attention
4	Strong influence	Decides strategy for use or development
5	Critical influence	Can "break" or "launch" the technology

Rating with AI experts: Ask 2–3 different models for a rating and take the average.

Example prompts for AI experts

Prompt #1 — Basic rating:

You are an expert in technology history and strategic analysis.
Rate on a scale from 1 to 5 how strongly the factor "[FACTOR NAME]" affected the development of the technology "[TECHNOLOGY NAME]" in the period [YEARS].

Rating criteria:
1 = almost no influence
3 = moderate influence, considered in planning
5 = critical influence, can decide success or failure

Give a short reason (1–2 sentences).

Prompt #2 — Comparative rating:

Compare two factors:

1) "[Factor A]"
2) "[Factor B]"

in the context of the development of [technology] in [period].

Which one had more influence? Why? Rate each on a scale of 1–5 with a reason.

Prompt #3 — Check weights:

I am distributing factor weights for PEST-analysis of [technology].
Check my distribution and suggest changes:

[INSERT TABLE WITH WEIGHTS]

Explain which factors deserve more/less weight and why. The sum of weights should stay ≈1.0.

Example table for expert ratings:

Factor	**Expert 1 (Grok)**	**Expert 2 (Qwen)**	**Expert 3 (ChatGPT)**	**Average (B)**
Governments use radio for military control	5	5	4	4.7
Possible for Info Interception	4	3	4	3.7
Investment in communication infrastructure	3	3	2	2.7
Growth of military technology industry	4	5	4	4.3
New role of information in war	5	5	5	5.0
Faster communication between ships	4	4	3	3.7
Marconi ship sets	2	2	3	2.3
Coastal wireless stations	2	3	2	2.3

STEP 3: Calculate the total score

Formula:

`Total Score = Factor Weight (A) × Strength of Influence (B)`

Example calculation:

Factor	**Weight (A)**	**Strength (B)**	**Total Score (A×B)**
Governments use radio for military control	0.20	4.7	0.94

Factor	Weight (A)	Strength (B)	Total Score (A×B)
Possible for Info Interception	0.10	3.7	0.37
Investment in communication infrastructure	0.05	2.7	0.14
Growth of military technology industry	0.15	4.3	0.65
New role of information in war	0.25	5.0	1.25
Faster communication between ships	0.15	3.7	0.56
Marconi ship sets	0.05	2.3	0.12
Coastal wireless stations	0.05	2.3	0.12
TOTAL	**1.00**	**—**	**≈4.15**

Interpretation of the sum: The higher the total score, the stronger the external environment affects the technology. Compare different technologies using this number.

STEP 4: Find positive and negative factors

Split factors into two groups by their type of influence:

🟢 **Positive (Opportunities)** — factors that help development:

- Governments use radio for military control
- Investment in communication infrastructure
- Growth of military technology industry
- New role of information in war
- Faster communication between ships
- Marconi ship sets
- Coastal wireless stations

🔴 **Negative (Threats)** — factors that create risks:

- Possible for Info Interception

Important: One factor can be both an opportunity and a threat at the same time (for example, military use = demand, but also a secrecy risk). In this case, write the row twice with different signs.

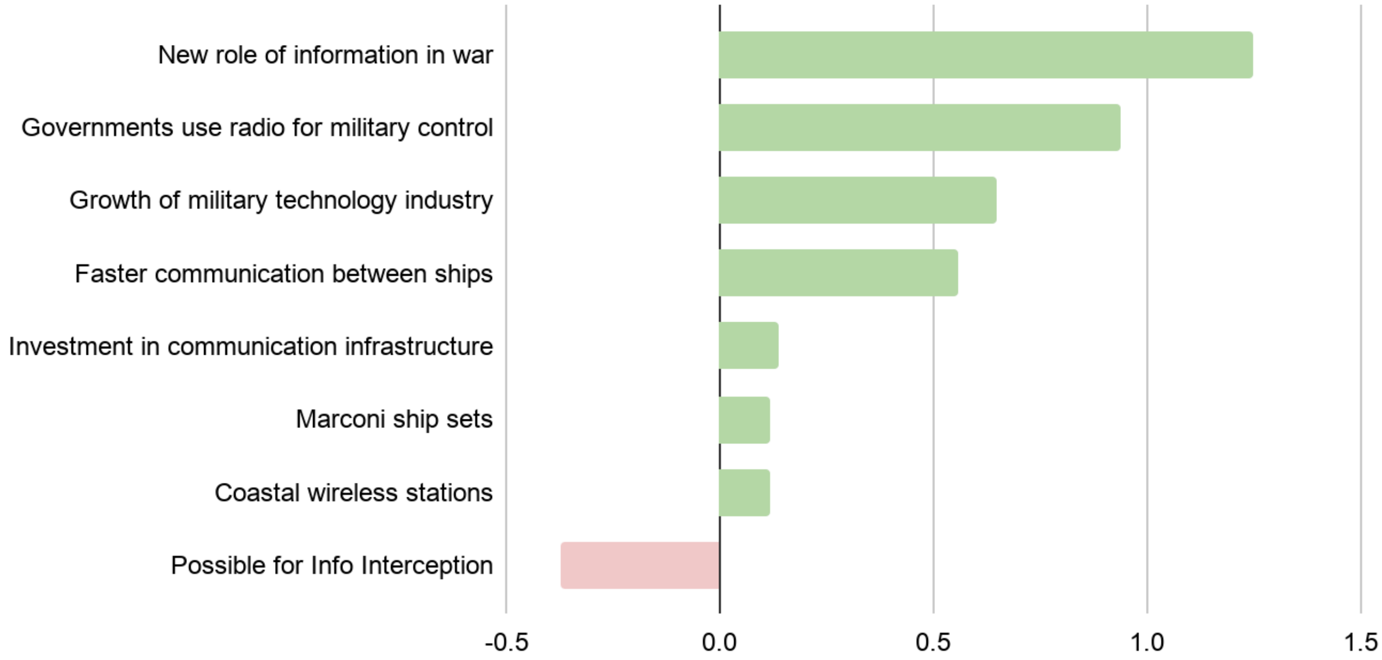
STEP 5: Show results with a chart

FINAL SUMMARY TABLE

Category / Factor	Weight (A)	Strength (B)	A×B	+/-	Comment
POLITICAL					
Governments use radio for military control	0.20	4.7	0.94	🟢	Key driver of demand and government support
Possible for Info Interception	0.10	3.7	0.37	🔴	Encourages encryption development, but brings risks
ECONOMIC					
Investment in communication infrastructure	0.05	2.7	0.14	🟢	Supporting factor, not deciding
Growth of military technology industry	0.15	4.3	0.65	🟢	Creates a stable market
SOCIAL					
New role of information in war	0.25	5.0	1.25	🟢	Changes the paradigm: information = strategic resource
Faster communication between ships	0.15	3.7	0.56	🟢	Practical value for navy
TECHNOLOGICAL					
Marconi ship sets	0.05	2.3	0.12	🟢	Local solution, does not change the main path
Coastal wireless stations	0.05	2.3	0.12	🟢	Infrastructure element, grows with demand
TOTAL	**1.00**	**—**	**4.15**		

Make a simple bar chart:

- **X-axis:** Factors (you can group them by PEST categories)
- **Y-axis:** Total Score (A×B)
- **Color:** 🟢 positive / 🔴 negative



This will clearly show which factors are most important and where to focus attention!

STEP 6: Interpretation and conclusions

Answer 3 key questions:

Question	Example
Which 2–3 factors need constant monitoring?	Example: "New role of information in war" (1.25) and "Governments use radio for military control" (0.94)
Which opportunities can we use now?	Growth in military orders → Creates a stable market where you can earn money fast.
Which risks should we fix first?	"Info Interception" → develop basic encryption methods

🚫 Self-check list

Mistake	How to avoid
❌ Mixing external and internal factors	Ask: "Can the project team change this factor?" If yes — it is not PEST
❌ Giving all factors the same weight	Use the checklist: "Does it affect availability? Can it block progress? How wide is its effect?"
❌ Rating influence without anchors	Check the 1–5 table and examples
❌ Ignoring negative factors	Even one 🔴 with a high score can be critical
❌ Doing analysis "in a vacuum"	Always link factors to a specific period and context